

Lab 7 Report: Autonomous Navigation with Nav2 and Multi-Waypoint Mission Planning

Objective

The objective of this lab was to perform autonomous navigation using the ROS 2 Nav2 stack, a pre-built occupancy map, AMCL localization, and TurtleBot3 in a Gazebo simulation environment. The lab also involved executing multi-waypoint missions and observing navigation-related topics and recovery behaviors.

Step 1: Launch Gazebo Simulation

The TurtleBot3 Gazebo world was launched using the TurtleBot3 simulation package. After the environment loaded successfully, the robot appeared in the simulated world. The node connections were verified using `rqt_graph`.

Observation

The robot spawned correctly in the simulation environment. All required simulation nodes were active and communicating properly.

Step 2: Launch Nav2 with Saved Map

The Nav2 navigation stack was launched using the previously generated map from Lab 5. The Map Server loaded the map successfully, while the Planner Server, Controller Server, BT Navigator, and AMCL localization node started without errors. RViz was launched for visualization.

Observation

The occupancy map was displayed correctly in RViz, and all navigation components were active. The robot was ready for localization and navigation tasks.

Step 3: AMCL Localization

Using the 2D Pose Estimate tool in RViz, the robot's initial position was set according to its location in Gazebo. The robot was moved slightly using teleoperation to improve localization accuracy.

Observation

The AMCL particle cloud gradually converged around the robot's actual position. The laser scan aligned accurately with the map walls, indicating successful localization.

Step 4: Single Goal Navigation

Three navigation goals were sent using the Nav2 Goal tool in RViz. For each goal, Nav2 generated a global path and the robot navigated autonomously to the target location.

Observation

The robot successfully reached all navigation goals. The global planner generated valid paths, while the local planner continuously adjusted the robot's trajectory to follow the path smoothly.

Step 5: Multi-Waypoint Navigation

A Python node (`waypoint_navigator.py`) was created using the FollowWaypoints action interface. Five waypoints were defined and sent to the Nav2 Waypoint Follower.

Observation

The robot successfully visited each waypoint in sequence. At every waypoint, a new path segment was generated automatically. The robot completed the entire mission and returned to the final destination without manual intervention.

Step 6: Visualization of Navigation Data

Several RViz visualization plugins were enabled, including Map, Global Costmap, Local Costmap, Global Path, Local Path, LaserScan, Odometry, TF, and AMCL Particle Cloud.

Observation

The Global Costmap represented the complete environment for path planning, while the Local Costmap continuously updated around the robot for obstacle avoidance. The global and local paths were clearly visible during navigation.

Task 4: Costmap Observation

The robot was driven near obstacles to observe costmap behavior.

Observation

The Local Costmap inflated around nearby obstacles, creating safety margins that prevented collisions. The Global Costmap remained relatively static and was used for long-range path planning.

Task 5: Recovery Behaviors

A dynamic obstacle was introduced into the robot's planned path inside Gazebo.

Observation

When the path became blocked, Nav2 triggered recovery behaviors. The robot paused, attempted replanning, and generated an alternative route around the obstacle. In some cases, the robot rotated in place before recalculating the path.

Conclusion

This lab successfully demonstrated autonomous navigation using the ROS 2 Nav2 stack. The integration of map loading, AMCL localization, path planning, waypoint following, and recovery behaviors provided a complete navigation workflow. Compared to Lab 5, which focused on map creation using SLAM, this lab emphasized utilizing an existing map for autonomous movement and mission execution. The experience provided practical understanding of modern robotic navigation systems and their real-world applications.